

The power of few key inventors

Importance and relevance

Peripheral regions often lose their best inventors to big cities (Verginer & Riccaboni, 2021). When they leave, the region also loses the weak-tie “bridges” that carry new ideas between groups that normally didn’t connect. Because moving into more complex technologies is path-dependent (step by step from the current skills) and most advanced tech is in big cities, losing these connectors makes capability upgrading harder and slows long-run growth (Agrawal et al. 2011; Mewes & Broekel, 2020).

Previous papers

Studies shows that top inventors move a lot and react to taxes. Studies finds that higher top tax rates push “superstar” inventors to choose other countries. In response, governments try to keep or attract these people using tools like tax breaks, special visas and active recruiting (Moretti & Wilson, 2017; Akcigit et al., 2016).

Moretti in 2021 found that after moving to a bigger hub, an inventor produces significantly more patents and receives more citations than before. The boost appears causally linked to the richer environment. The findings align with the idea that large clusters offer thicker labor markets, complementary expertise and knowledge spillovers that enhance individual productivity.

Tóth and Lengyel (2021) find that firms achieve more high-impact innovations when they hire inventors who are brokers in the global co-inventor network. Personally i think this firm-level result could echoe at the regional scale as well.

Zucker et al. (2007) report that typically one or two star scientists keep their university roles while founding or co-founding a new biotech firm. The less often scenario is they become involved with an existing firm. The paper showed a mechanism: even few star scientists can found new firms or pull existing firms into a new field, because they bridge university-industry networks and attract people, money and ideas.

Niche & Question

We still don’t know whether keeping or bringing just a few well-connected inventors can truly change a region’s path into a more advanced tech. And if yes how many are needed, what kind of connectors needed, where to place them and how fast the effect appears.

Question: Compared with similar regions does keeping or attracting some “connector” inventors increase the chance that the region enters at least one more complex technology within a few years? (of course while accounting for example region size, education, R&D spending)

Stakeholders

Regional agencies, immigration/tax offices, R&D-heavy companies and universities could use this result to choose how to act. They can either spend on big, broad incentives for many people or run targeted programs. This could be for example special visas for top researchers, help with moving costs or shared jobs between a university and a company. The goal is to bring in or keep a few “connector” inventors who link local teams to cutting-edge know-how.

Citations

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